

##CRISCd - Information Technology and Security Principles

###Objectives:

At the end of this episode, I will be able to understand:

1. Encryption is a Mathematical means of altering data from readable forms to unreadable forms (ciphertext)
2. Data encryption uses two forms:
 - Symmetric
 - Asymmetric
3. Used primarily for confidentiality
4. Symmetric Key Cryptography:
 - Most common system in place is Advanced Encryption Standard (AES)
 - Shorter keys are easier to crack
 - Shares the same secret key
 - Disadvantages:
 - One party delivers keys (SPOF)
 - No way to tell who originated message
5. Asymmetric Key Cryptography:
 - Diffie and Hellman created a different method called the Diffie-Hellman model
 - Two related keys are created: Private and Public
 - Public keys are freely distributed. Private keys are only held by the creators and receivers
 - Disadvantages:
 - Computer resource intensive
 - Much slower than Symmetric
6. Digital Signatures combines a hash function with the ability of PKI to prove the author's identity
7. Creates excellent nonrepudiation
8. Digital signatures do NOT encrypt the message...just the digest
9. PKI uses Certificates to verify the identity of the owner of a public key
10. Symmetric keys are excellent for confidentiality but difficult to manage
11. Asymmetric algorithms are much slower but much more secure
12. Using both is the secret
13. Hashing algorithms can help make sure of message integrity
14. Certificates provide assurance that a public key belongs to the right owner

Additional resources used during the episode can be obtained using the download link on the overview episode.

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

<http://www.isaca.org>